

Instructions

This mock mirrors the Tredence data-science assessment: aptitude, verbal, statistics, ML and CS MCQs plus SQL and Python.

Attempt in one ~2-hour sitting. This is Mock 2 of 3; each uses a different question set.

Section 1 - Aptitude & Reasoning

Q1. 250 is what percent of 1000?

(A) 20%

(B) 25% ✓

(C) 30%

(D) 15%

Answer: (B) 25%

Authenticity: Community Reported

Q2. If $A:B = 3:5$ and $A = 27$, then B is:

(A) 40

(B) 45 ✓

(C) 50

(D) 36

Answer: (B) 45

Authenticity: Community Reported

Q3. If 5 machines make 5 units in 5 min, 100 machines make 100 units in:

(A) 100 min

(B) 5 min ✓

(C) 20 min

(D) 1 min

Answer: (B) 5 min

Authenticity: Frequently Repeated

Q4. Next term: 2, 4, 8, 16, ?

(A) 24

(B) 32 ✓

(C) 20

(D) 30

Answer: (B) 32

Authenticity: Frequently Repeated

Q5. Odd one out: 8, 27, 64, 100.

- (A) 8
- (B) 27
- (C) 64

(D) 100 ✓

Answer: (D) 100

Authenticity: Community Reported

Q6. All roses are flowers. Some flowers fade. Conclusion: Some roses fade.

- (A) Follows

(B) Does not follow ✓

- (C) Always true
- (D) None

Answer: (B) Does not follow

Authenticity: Frequently Repeated

Section 2 - Verbal

Q1. Synonym of CONCISE:

- (A) Lengthy

(B) Brief ✓

- (C) Vague
- (D) Complex

Answer: (B) Brief

Authenticity: Community Reported

Q2. Neither of the two answers ____ correct.

- (A) are

(B) is ✓

- (C) were
- (D) have

Answer: (B) is

Authenticity: Community Reported

Q3. (Same passage) Risk highlighted:

- (A) Too much intuition

(B) Poor data quality leading to misleading conclusions ✓

- (C) Too many employees
- (D) Excessive testing

Answer: (B) Poor data quality leading to misleading conclusions

Authenticity: Research-Backed Company Style

Section 3 - Statistics & Probability

Q1. Find the median of 3, 1, 4, 1, 5, 9, 2.

(A) 2

(B) 3 ✓

(C) 4

(D) 1

Answer: (B) 3

Authenticity: Community Reported

Q2. Variance equals:

(A) The standard deviation

(B) The square root of the SD

(C) The square of the standard deviation ✓

(D) The mean of deviations

Answer: (C) The square of the standard deviation

Authenticity: Community Reported

Q3. The median corresponds to which percentile?

(A) 25th

(B) 50th ✓

(C) 75th

(D) 100th

Answer: (B) 50th

Authenticity: Community Reported

Q4. For mutually exclusive events $P(A)=0.2$, $P(B)=0.3$, $P(A \text{ or } B)$ is:

(A) 0.06

(B) 0.5 ✓

(C) 0.1

(D) 0.25

Answer: (B) 0.5

Authenticity: Community Reported

Q5. If A and B are independent, $P(A | B)$ equals:

(A) $P(B)$

(B) $P(A)$ ✓

(C) $P(A \text{ and } B)$

(D) 0

Answer: (B) $P(A)$

Authenticity: Frequently Repeated

Q6. Probability a drawn card is a heart:

(A) $1/13$

(B) $1/4$ ✓

(C) $1/52$

(D) $1/2$

Answer: (B) $1/4$

Authenticity: Community Reported

Q7. A Type I error is:

(A) Failing to reject a false null

(B) Rejecting a true null hypothesis ✓

(C) Accepting correctly

(D) A sampling error

Answer: (B) Rejecting a true null hypothesis

Authenticity: Frequently Repeated

Q8. For a normal distribution, ~what percent lies within one SD of the mean?

(A) 50%

(B) 68% ✓

(C) 95%

(D) 99.7%

Answer: (B) 68%

Authenticity: Community Reported

Q9. A correlation coefficient of 0 indicates:

(A) Perfect positive

(B) Perfect negative

(C) No linear relationship ✓

(D) Causation

Answer: (C) No linear relationship

Authenticity: Community Reported

Section 4 - Machine Learning

Q1. Which technique reduces overfitting?

(A) Increasing complexity

(B) Regularization ✓

(C) Removing validation

(D) Fewer epochs only

Answer: (B) Regularization

Authenticity: Frequently Repeated

Q2. In the bias-variance tradeoff, high bias leads to:

(A) Overfitting

(B) Underfitting ✓

(C) Perfect fit

(D) Data leakage

Answer: (B) Underfitting

Authenticity: Community Reported

Q3. Which is NOT a supervised algorithm?

- (A) Linear Regression
- (B) Logistic Regression

(C) K-means ✓

- (D) Random Forest

Answer: (C) K-means

Authenticity: Frequently Repeated

Q4. Precision is:

- (A) $TP/(TP+FN)$

(B) $TP/(TP+FP)$ ✓

- (C) $TN/(TN+FP)$

- (D) $TP/Total$

Answer: (B) $TP/(TP+FP)$

Authenticity: Frequently Repeated

Q5. Cross-validation is used to:

- (A) Increase data size

(B) Estimate generalization to unseen data ✓

- (C) Reduce classes

- (D) Encode features

Answer: (B) Estimate generalization to unseen data

Authenticity: Community Reported

Q6. Typical loss for binary classification:

- (A) MSE

(B) Binary cross-entropy ✓

- (C) MAE

- (D) R-squared

Answer: (B) Binary cross-entropy

Authenticity: Community Reported

Q7. The sigmoid function outputs values in:

- (A) -1 to 1

(B) 0 to 1 ✓

- (C) 0 to inf

- (D) -inf to inf

Answer: (B) 0 to 1

Authenticity: Community Reported

Q8. If the learning rate is too high, training may:

- (A) Converge faster with no downside

(B) Fail to converge and overshoot ✓

- (C) Always reach global minimum

- (D) Stop updating

Answer: (B) Fail to converge and overshoot

Authenticity: Community Reported

Q9. L1 (Lasso) regularization tends to produce:

(A) Dense weights

(B) Sparse models (some weights zero) ✓

(C) Larger weights

(D) No change

Answer: (B) Sparse models (some weights zero)

Authenticity: Community Reported

Section 5 - CS Fundamentals

Q1. Which data structure follows LIFO?

(A) Queue

(B) Stack ✓

(C) Array

(D) Heap

Answer: (B) Stack

Authenticity: Community Reported

Q2. Inorder traversal of a BST gives:

(A) Reverse order

(B) Sorted order ✓

(C) Random order

(D) Level order

Answer: (B) Sorted order

Authenticity: Community Reported

Section 6 - SQL

Problem: Count Employees per Department

Count employees per department, ordered by count desc then dept asc.

```
SELECT dept, COUNT(*) AS emp_count
FROM emp
GROUP BY dept
ORDER BY emp_count DESC, dept ASC;
```

Expected Output

```
dept|emp_count
Tech|3
Sales|2
HR|1
```

Section 7 - Python

Problem 1: Find the Missing Number

Difficulty: *Easy* | Pattern: *Math / Summation* | Authenticity: *Community Reported*

Problem Statement

An array holds n distinct numbers from $0..n$ with exactly one missing. Return it.

Input Format

Line 1: n . Line 2: the n numbers.

Output Format

The missing number.

Constraints

$1 \leq n \leq 10^6$

Sample Input

```
8
3 0 1 5 2 7 6 8
```

Sample Output

```
4
```

Python Solution

```
def missing_number(nums):
    n = len(nums)
    return n * (n + 1) // 2 - sum(nums)

print(missing_number([3, 0, 1, 5, 2, 7, 6, 8]))
```

Step-by-Step Explanation

Sum of $0..n$ minus the actual sum gives the missing value. $36 - 32 = 4$.

Dry Run

expected 36, actual 32 -> 4

Time Complexity: **$O(N)$** | Space Complexity: **$O(1)$**

Tredence

Full-Length Mock Test 2

Applied Role(s):

Analyst - Data Scientist

Online Assessment Preparation Material

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Every question carries an authenticity label. See the legend on the next page.

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Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.